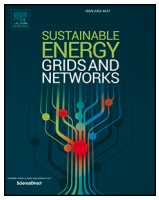
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A physics-constrained zonal pseudo-measurement imputation method for active distribution networks under advanced metering infrastructure data missingness

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ABSTRACT

Fine-grained monitoring of active distribution networks relies on complete measurements at the distribution-transformer level; however, owing to communication-link latency the advanced metering infrastructure (AMI) suffers delayed data arrival, so that random missingness is pervasive across sampling cross-sections, while purely data-driven imputation struggles to guarantee physical consistency, so its outputs cannot directly serve feasibility-sensitive downstream applications. To address this, we propose a Physics-Informed Gated Attention Transformer, abbreviated PIGAT, which performs zonal pseudo-measurement imputation following a "correct-then-propagate" principle: the network is divided into radial sub-zones with tie switches as boundaries, the boundary tie branches are observed by SCADA, and the outgoing tie power flows act as free variables that characterize inter-zone coupling. The framework consists of three mutually coupled mechanisms. First, a confidence-aware gate corrects missing entries before spatial aggregation to suppress missing-value contamination, and a physics-informed impedance attention bias encodes electrical proximity by cumulative branch impedance. Second, a differentiable DistFlow hard-constraint output head with denoising enhancement embeds the branch power-flow equations as an analytical recursion, so that the imputation result strictly satisfies this physical constraint by construction. Third, physical partitioning combined with per-node shared projection decouples the parameter count from the number of nodes N and reduces the per-zone node scale, balancing scalability and parallel-training efficiency. Under scenarios with different missing rates, noise levels, and DER penetrations, zonal pseudo-measurement imputation experiments are conducted on the modified IEEE 33-/123-bus systems, validating the synergistic advantages of the proposed method in accuracy, physical consistency, and scalability.

1. Introduction

The carbon-neutrality-oriented energy transition is integrating distributed energy resources (DERs) such as rooftop photovoltaics, small wind turbines, and electric-vehicle charging stations into distribution networks, turning them from unidirectional passive systems into bidirectional active ones with strong randomness and pronounced planning uncertainty [1]. Accurate imputation of missing measurements at the distribution-transformer (DT) level, i.e. high-quality pseudo-measurements, is therefore a key enabler for distribution system state estimation (DSSE) [2] and for the functions it feeds: the demand-response optimization of Fotopoulou et al. [3] smooths the load curve at transformer and substation level to mitigate voltage drops and congestion, and can be dispatched no more accurately than the DT-level measurements it consumes. Inaccurate pseudo-measurements mean biased estimation, suboptimal dispatch, excessive reserve, and reduced renewable integration, a dependence equally reported for load forecasting [4].

Unlike system-level state estimation, which benefits from aggregation-induced smoothing, DT-level imputation faces three challenges. First, sparsity and incompleteness: real-time SCADA covers only substations and a few feeder heads, whereas most DT nodes rely on AMI smart meters sampled every 15 min [5]; under wireless/PLC latency and congestion only about 70% of the data scheduled on the hour arrive at the aggregation cutoff and the rest trickle in over several sampling periods, leaving random missingness that column-wise imputation can

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Nomenclature

Sets, indices and dimensions

$G = (V, \mathscr{E})$	Network graph with node set V and branch set $\mathscr{E}$
$Va,\ \mathscr{E}a$	Node and branch set of zone a
$\pi(k)$	Parent node of node k
$C(k)$	Downstream node set of node k
out(k)	Set of tie branches leaving node k
$P(i,j)$	Unique path between nodes i and j in a radial zone
$\mathscr{F}free,\ \mathscr{F}dep$	Free and dependent output-channel set
N, K	Number of nodes and of branches of a zone
N_a	Number of zones

Electrical quantities

P_i, Q_i	Active and reactive power injection at node i (p.u.; kW, kvar)
$\widetilde{P_k}$, $\widetilde{Q_k}$	Effective injection at node k, outgoing tie flows included (p.u.; kW, kvar)
P_{ik}, Q_{ik}	Sending-end active and reactive flow of branch (i→k) (p.u.; kW, kvar)
P_f, Q_f	Active and reactive flow of a tie-switch branch; the further subscripts out,e and entry denote an outgoing branch e and the entry branch of a zone (p.u.; kW, kvar)
V_i	Voltage magnitude at node i (p.u.; kV)
V_{root}	Voltage magnitude at the root node of a zone (p.u.; kV)
$\lvert I_f\rvert$	Current magnitude of tie-switch branch f (p.u.; A)
ℓ_{ik}	Squared current magnitude of branch (i→k) (p.u.; A^2)
r_{ik}, x_{ik}	Resistance and reactance of branch (i→k) (p.u.; Ω)
$\lvert Z_{uv}\rvert$	Impedance magnitude of branch (u,v) (p.u.; Ω)
$d_{\text{elec}}(i,j)$	Electrical distance between nodes i and j (p.u.; Ω)
σ_V, σ_S	Noise standard deviation of the voltage and of the power channels (% of full scale)

Model variables

$\mathbf{X}$	Feature tensor of a sliding window (p.u.)
$\mathbf{M}$	Missingness mask tensor
$\mathbf{x}_{\text{mark}}$	Time-mark tensor
$\mathbf{m}_{t,i}$	Mask embedding of node i at step t
$c_{t,i}$	Observation confidence of node i at step t
$\mathbf{g}_{t,i}$	Feature-level gating vector
$\mathbf{h}_{t,i}$	Projected hidden vector
$\widetilde{\mathbf{h}_{t,i}}$	Gated hidden vector
$\mathbf{h}'_{t,i}$	Fused hidden vector
$\mathbf{h}''_{t,i}$	PIAB output vector, after residual mixing and normalization
$\mathbf{e}_{t,i}$	Encoder output vector
γ_h	Strength of the impedance attention bias of head h
λ_h	Spatial scale of the impedance attention bias of head h (p.u.; Ω)
β_h	Offset of the impedance attention bias of head h
$\hat{\mathbf{y}}_t^{\text{f}}$	Free-variable vector predicted at step t, before temporal denoising (p.u.)
$\tilde{\mathbf{y}}_t^{\text{f}}$	Free-variable vector after temporal denoising (p.u.)
$\hat{\mathbf{y}}_t^{\text{d}}$	Dependent-variable vector returned by f_{DF} (p.u.)
$\hat{\mathbf{y}}_t$	Complete output vector stacking the free, dependent and auxiliary channels (p.u.)
α_a	Residual weight of the PIAB branch
α_d	Residual-mix weight of the denoising branch
f_{DF}	Differentiable DistFlow operator
$\mathbf{J}_{f\text{DF}}$	Jacobian of f_{DF}
θ	Learnable parameters of the model
$\mathscr{L}$	Training loss

Abbreviations

AMI	Advanced metering infrastructure
DT	Distribution transformer
DER	Distributed energy resource
SCADA	Supervisory control and data acquisition
DSSE	Distribution system state estimation
GNN	Graph neural network
BiGRU	Bidirectional gated recurrent unit
PIGAT	Proposed physics-informed gated attention transformer
PIAB	Impedance attention bias of PIGAT
DHC	Denoising DistFlow hard-constraint output head of PIGAT

hardly handle [6],an uncertainty that remains open for deep-learning DSSE [7]. Second, physical self-consistency: Volt/VAR optimization and emergency dispatch need pseudo-measurements satisfying the power-balance and voltage-drop relations of radial networks, which purely data-driven methods cannot guarantee however accurate [8]. Third, scale under limited computing resources: practical networks contain hundreds to thousands of buses, and a global model whose feature dimension grows quadratically with the node number readily triggers dimensional explosion [9].

In data-driven pseudo-measurement modeling, traditional methods rely on typical load-profile scaling or shallow neural-network regression, as in the estimator of Manitsas et al. [10], and can hardly capture complex spatiotemporal correlations; recent work adds temporal structure, Wang et al. [11] attaching attention to a recurrent network and Qiu et al. [12] combining CNN-BiLSTM with a customized genetic algorithm. Accuracy improves, yet none resolves the three gaps below jointly.

In missing-data handling, MissForest [13] imputes iteratively with random forests; BRITS [14] propagates bidirectionally in a recurrent network; GAIN [15] imputes adversarially; SAITS [16] jointly optimizes reconstruction and imputation by self-attention; GRIN [17] performs bidirectional message passing on a fixed graph; and GRU-D [18] shows that the missingness pattern itself carries discriminative information. Within power systems, Donti et al. [19] complete missing voltage entries by matrix completion under low observability, which is effective but static and cannot reproduce nonlinear dynamics. In spatial encoding, the survey of Jin et al. [20] documents graph neural networks (GNNs) for time-series imputation, Zhuang et al. [21] build a multi-scale spatio-temporal graph network for multi-energy load forecasting, Defferrard et al. [22] approximate spectral filters by Chebyshev polynomials, and Wu et al. [23] unroll a spatiotemporal graph convolutional network for DSSE, all of them topology-encoding frameworks; Graphormer [24] uses the shortest-path distance as a learnable attention bias, and ALiBi [25] introduces a linear positional bias into language-model attention. These architectures share two intertwined limitations: they apply graph message passing directly to the raw input without pre-correcting missing entries, so placeholder zeros propagate through aggregation into neighboring representations and cause *missing-value contamination*; and their fixed or binary-weighted adjacency matrices impose uniform coupling on all connected nodes, ignoring the pronounced heterogeneity of branch impedance. Although the physics-informed GNN of Ngo et al. [26] alleviates the spatial-fidelity issue, it does not address missing data. This constitutes Research Gap 1: existing spatiotemporal models fail to jointly cope with the random missingness induced by AMI communication latency and the physics-agnostic nature of spatial encoders, a dual difficulty.

Beyond spatiotemporal modeling, even high imputation accuracy yields limited operational value if it violates the underlying physical laws [8]. In the optimization domain, DC3 [27] proposes a differentiable

Table I
Comparison of representative methods under the three research gaps.

Method	Task	Missing correction	Physics-informed spatial encoding	Physical constraint	Scalable architecture
SAITS [16]	Imputation	✓	×	×	×
GRIN [17]	Imputation	✓	△ fixed-adjacency GCN	×	×
Graphormer [24]	Graph representation	×	✓ SPD bias	×	×
ALiBi [25]	NLP	×	△ linear positional bias	×	×
ANN-PM [10]	Pseudo-measurement	×	×	×	×
CNN-BiLSTM [12]	Pseudo-measurement	×	×	×	×
PI-GNN [26]	State estimation	×	✓ topology-aware	✓ soft penalty	×
PhysNet [30]	State estimation	×	△ separable NN	×	△ separable architecture
DC3 [27]	OPF	×	×	✓ hard constraint	×
PIGAT (this work)	Pseudo-measurement imputation	✓ confidence gating	✓ PIAB (electrical distance)	✓ DistFlow (hard)	✓ partitioning + dimension decoupling

completion-correction framework to enforce AC-OPF equality constraints. In state estimation, Ngo et al. [26] incorporate branch-flow and current equations into a topology-aware GNN as a soft constraint, which cannot guarantee strict feasibility and is sensitive to the weighting of the physical term. The DistFlow branch power-flow model of Baran and Wu [28] describes radial topologies rigorously, and the linearized variant of Yang et al. [29] is widely applied in reactive-power optimization. Introducing hard constraints into pseudo-measurement imputation under missing data nevertheless remains open. This constitutes Research Gap 2: pseudo-measurement imputation lacks hard physical constraints, so that its outputs cannot directly serve the downstream applications that are sensitive to physical feasibility.

Even with an improved spatiotemporal encoder and a physics-constrained output head, deployment on a large-scale distribution network still faces scalability challenges. When the features of N nodes are concatenated into a single input vector, the model dimension $d_{\text{model}}=N\times H$ (with H the per-node hidden dimension) makes the parameter count of each projection matrix scale as $\mathcal{O}(N^2H^2)$, a cost prohibitive for networks with hundreds to thousands of buses. Zamzam and Sidiropoulos [30] exploit the radial structure to make the measurement-to-state mapping separable and so reduce the parameter count, but do not preserve the inter-zone power coupling that zonal modeling needs. Electrical-distance partitioning has been applied to voltage control, as in the zonal scheme of Zhao et al. [9], which divides a network hosting distributed PV and regulates each zone independently; no existing work, however, integrates topology-aware physical partitioning with a parameter-efficient architecture so as to reduce order in the topological and the parameter dimensions at once. This constitutes Research Gap 3: existing research lacks a scalable, dimension-decoupled imputation architecture that can control computational complexity through physical partitioning while preserving inter-zone power--coupling information.

Table I compares representative methods along the three threads above. None fills Gap 1 (missing-data correction and physics-informed spatial encoding), Gap 2 (hard physical constraints), and Gap 3 (scalability) at once, so resolving the three jointly remains open.

✓ = fully supported; △ = partially supported; × = not supported.

To close the three gaps, this paper proposes the Physics-Informed Gated Attention Transformer (PIGAT), whose architecture follows a "correct-then-propagate" principle: missing-data correction precedes physics-informed spatial encoding, which precedes temporal modeling and the physics-constrained output. Confidence-aware gating with a physics-informed impedance attention bias (PIAB) addresses Gap 1: feature-level gating modulated by observation-density confidence pre-corrects missing entries before spatial aggregation, and impedance-biased attention replaces spectral graph convolution, generalizing node distance from topological hops to a cumulative branch-impedance electrical distance with a learnable per-head spatial scale. A differentiable DistFlow hard-constraint output head with denoising (DHC) addresses Gap 2: only the free variables (root voltage, nodal injections, outgoing tie flows) are predicted, the dependent variables following analytically from leaf-to-root power accumulation and root-to-leaf voltage sweeping and so satisfying the branch power-flow equations by construction; a per-channel temporal convolution suppresses noise beforehand, and the objective reduces to a single Huber loss confining the physical-violation degree (PVD) to 10^{-5} p.u. Zone-independent modeling with dimension decoupling addresses Gap 3: tie switches partition the topology into independent radial sub-zones with outgoing tie flows as free variables, and per-node shared projection makes the parameter count depend only on H rather than N, jointly yielding a 3.5–4.0 × parallel-training speedup.

Section 2 introduces the graph-modeling and multi-source-data framework; Section 3 elaborates the method; Section 4 reports validation on the modified IEEE 33- and 123-bus systems; Section 5 discusses strengths, limitations and future work; and Section 6 concludes.

2. Distribution network graph modeling and multi-source data definition

2.1. Topological graph model

A network is modeled as a graph $\mathcal{G}=(\mathcal{V},\mathcal{E})$ with node set $\mathcal{V}$ and branch set $\mathcal{E}$. Its adjacency matrix $\mathbf{A}\in\mathbb{R}^{|\mathcal{V}|\times|\mathcal{V}|}$ satisfies $a_{ij}=1$ when nodes i and j are physically connected and 0 otherwise. Each branch $(u, v)\in\mathcal{E}$ carries a known resistance r_{uv}, reactance x_{uv}, and impedance magnitude $|Z_{uv}|=\sqrt{r_{uv}^2+x_{uv}^2}$, which form a static topological prior.

2.2. Multi-source measurement deployment

Two heterogeneous measurement types are deployed jointly, a configuration that the power-flow model of a medium-voltage network must itself account for [31].

(i) **Tie-switch SCADA**: placed only at feeder tie switches, measuring P_f, Q_f, $|I_f|$, and the one-end voltage magnitude $|V|$ at a second-level rate, transmitted via RTUs over a dedicated network and serving as the boundary condition of each zone.

(ii) **In-zone AMI**: each DT node carries a smart meter measuring the injections (P_i, Q_i) and voltage magnitude V_i every 15 min; under wireless/PLC latency part of these data miss the aggregation cutoff, leaving each cross-section with random missingness (typically 20%-60%) plus sampling noise.

2.3. Measurement tensor and missingness mask

Given a batch size B, window length T, zone node number N, and branch number K, the input consists of three tensors.

(i) **Feature tensor** $\mathbf{X}\in\mathbb{R}^{B\times T\times D}$: a cross-section $\mathbf{X}_t=[\mathbf{X}_t^{\text{node}},\mathbf{X}_t^{\text{edge}}]$ concatenates nodal and edge measurements, $\mathbf{X}_t^{\text{node}}\in\mathbb{R}^{N\times 3}$

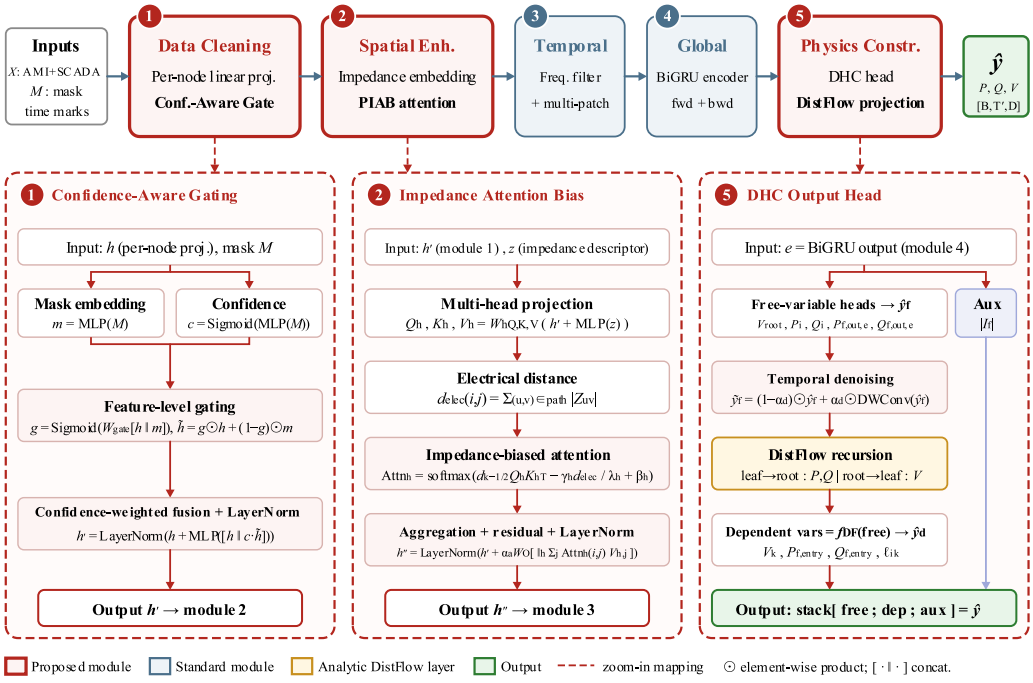


Fig. 1. Overall architecture of the PIGAT model.

holding (P, Q, V) and $\mathbf{X}_t^{\text{edge}} \in \mathbb{R}^{K\times 6}$ the edge measurements (P_f, Q_f, $|I_f|$) with the static parameters (r, x, $|Z|$); hence $D = 3N + 6K$.

(ii) **Missingness-mask tensor** $\mathbf{M} \in \{0,1\}^{B\times T\times D}$: of the same shape as $\mathbf{X}$, it records the status of each feature column ($m_{t,l}$=0 missing, 1 valid). Missingness and noise are injected per column: SCADA and static-parameter columns stay valid, while AMI columns are zeroed at random according to the specified missing rate.

(iii) **Time-mark tensor** $\mathbf{x}_{\text{mark}} \in \mathbb{R}^{B\times T\times 4}$: each timestamp provides four periodic features (hour of day, day of week, day of month, day of year), normalized to $[-0.5, 0.5]$:

$$x_{\text{mark}}^{(t,d)} = \frac{t_{t,d}}{t_{d,\max}} - 0.5 \tag{1}$$

where $t_{t,d}$ is the original value of the d-th temporal feature at step t and $t_{d,\max}$ is its maximum (e.g. 24 for the hour of day). The time mark is injected through an additive embedding, giving the model a periodicity inductive bias.

A sliding window cuts the three tensors into samples, each mini-batch holding B windows. Zero positions in $\mathbf{M}$ are corrected by the confidence-aware gate, updated by loss-driven gradients, and finally projected onto the DistFlow feasible set by DHC.

3. Methodology

3.1. Overall architecture

PIGAT cascades its stages by the "correct-then-propagate" principle: the gate precedes PIAB so spatial aggregation does not amplify the error of missing entries; PIAB precedes the BiGRU so per-node representations carry a physically informed topological context; and DHC comes last. Given $\mathbf{X}$, $\mathbf{M}$, and $\mathbf{x}_{\text{mark}}$ of Section 2.3, five modules are connected in series, matching the numbers ①-⑤ in Fig. 1: the gate outputs $\mathbf{h}'$ of Eq. (5), PIAB outputs $\mathbf{h}''$ of Eq. (8), the BiGRU outputs $\mathbf{e}$ of Eq. (11), and the DistFlow recursion of Module 5 outputs $\hat{\mathbf{y}}$. The three components in red, namely the gate (§3.3), PIAB (§3.4), and DHC (§3.7), are the core innovations, performing missing correction, physics-informed spatial encoding, and hard physical guarantee; PIAB and DHC inject physical priors at the feature and output layers, forming a dual-layer physical encoding. Partitioning and dimension decoupling (§3.2) run throughout, and the remaining stages are adapted standard modules.

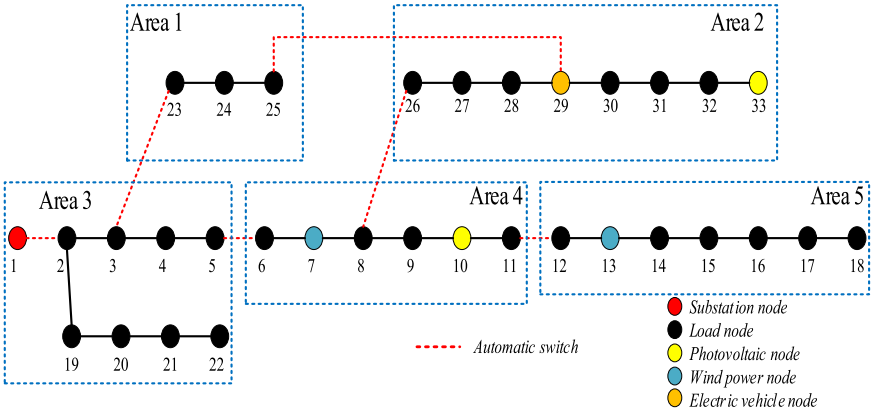


Fig. 2. Partitioning diagram of the modified IEEE 33-bus distribution network.

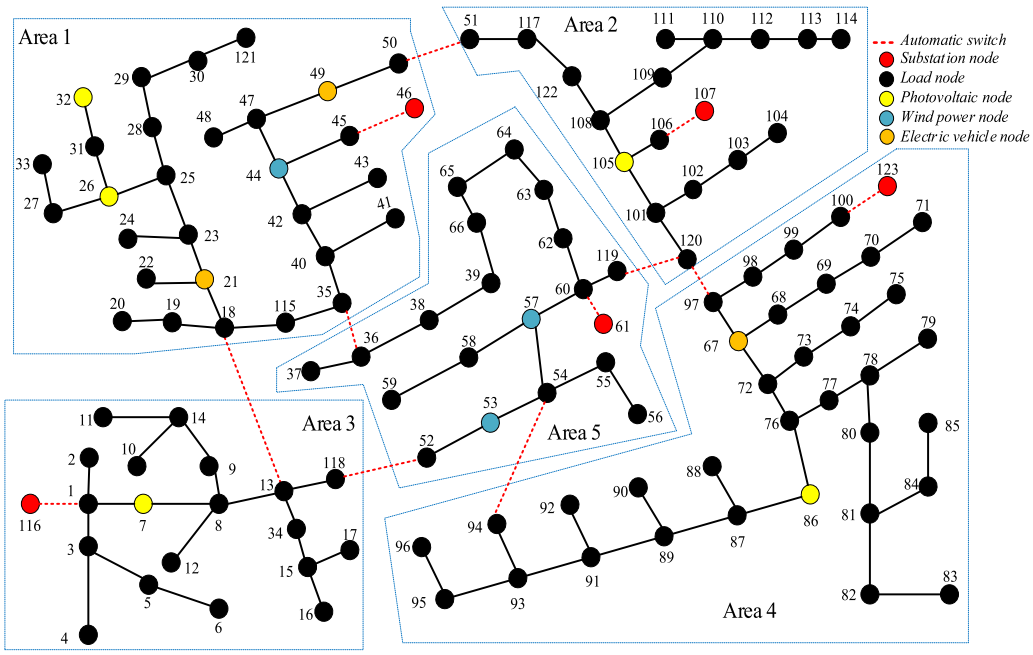


Fig. 3. Partitioning diagram of the modified IEEE 123-bus distribution network.

3.2. Physical partitioning and dimension decoupling

3.2.1. Tie-switch-based physical partitioning

Taking SCADA-equipped tie switches as boundaries, the topology is divided into N_a independent radial subgraphs $\mathcal{G}_a = (\mathcal{V}_a, \mathcal{E}_a)$; Figs. 2 and 3 show the five subgraphs of each test system. Each zone is fixed by the SCADA boundary condition at its root node (§2.2) and coupled to its neighbors by tie-branch power exchange, while in-zone AMI supplies node-level observations. Three advantages follow: each zone completes its DistFlow recursion independently; zone-independent training breaks the global coupling, so sub-zones run in parallel on the GPU; and partitioning confines N to a small scale, suppressing the quadratic complexity of spatial attention.

3.2.2. Dimension-decoupling strategy

If the features of a zone with N nodes were concatenated into one vector of length NH and projected as in a conventional temporal Transformer, each projection matrix would grow quadratically with N. PIGAT instead projects every node independently through a shared linear layer into $\mathbf{h}_{t,i} \in \mathbb{R}^H$, giving a hidden tensor in $\mathbb{R}^{B\times T\times N\times H}$ for all subsequent modules. Hence the parameter count depends only on H: one set of weights transfers across sub-zone sizes and capacity is tuned by H alone. PIAB self-attention remains $\mathcal{O}(N^2H)$: dimension decoupling removes the quadratic dependence in the *parameter* dimension, while §3.2.1 suppresses that in the *node* dimension by keeping the per-zone N small.

3.3. Confidence-aware gating

At each step t and bus i, the raw node features are projected into the per-node hidden space by a shared linear layer, and the edge features are added by broadcasting:

$$\mathbf{h}_{t,i} = \mathbf{W}_{\text{node}}\mathbf{x}_{t,i} + \text{GELU}\left(\mathbf{W}_{\text{edge}}\mathbf{x}_{\text{edge},t}\right) \in \mathbb{R}^H \tag{2}$$

where $\mathbf{x}_{t,i}$ is the raw feature vector of node i, $\mathbf{x}_{\text{edge},t}$ the edge feature vector, $\mathbf{W}_{\text{node}}$ and $\mathbf{W}_{\text{edge}}$ learnable weights, and GELU(·) the Gaussian error linear unit.

Under high missing rates many entries carry no information; fed into spatial attention, placeholder zeros propagate to neighbors and contaminate them. Following GRU-D [18], the gate uses three pathways: one perceptron embeds the binary mask, $\mathbf{m}_{t,i}$=MLP($\mathbf{M}_{t,i}$); a second returns a scalar confidence $c_{t,i}$=Sigmoid(MLP($\mathbf{M}_{t,i}$)) summarizing data availability; and a feature-level gate interpolates element-wise between the projected input and the mask embedding, each MLP(·) being a two-layer perceptron with its own weights:

$$\mathbf{g}_{t,i} = \text{Sigmoid}\left(\mathbf{W}_{\text{gate}}\left[\mathbf{h}_{t,i} \parallel \mathbf{m}_{t,i}\right]\right) \tag{3}$$

$$\widetilde{\mathbf{h}}_{t,i} = \mathbf{g}_{t,i} \odot \mathbf{h}_{t,i} + \left(1 - \mathbf{g}_{t,i}\right) \odot \mathbf{m}_{t,i} \tag{4}$$

where Sigmoid(·) is the logistic function, $[\cdot\|\cdot]$ denotes concatenation, $\odot$ the element-wise product, and $\mathbf{W}_{\text{gate}}$ the learnable gating weight. The gated representation is then normalized and fused by a confidence-weighted residual:

$$\mathbf{h}'_{t,i} = \text{LayerNorm}\left(\mathbf{h}_{t,i} + \text{MLP}\left(\left[\mathbf{h}_{t,i} \parallel c_{t,i}\cdot\widetilde{\mathbf{h}}_{t,i}\right]\right)\right) \tag{5}$$

where LayerNorm(·) is layer normalization. When confidence is high the fusion is dominated by the corrected representation $\widetilde{\mathbf{h}}_{t,i}$; when it is low that term vanishes and the raw projection drives the output. Sinusoidal positional encoding [32] then provides time-step ordering.

3.4. Impedance attention bias (PIAB)

ChebNet-style spectral convolutions weight all connected nodes by a fixed or binary adjacency and can hardly encode impedance heterogeneity; ALiBi [25] and Graphormer [24] bias attention by the sequence-position distance and the shortest-path hop count, the former ignoring graph structure, the latter treating distance as a discrete scalar. This section keeps the shortest-path metric of Graphormer but replaces the hop count with the cumulative branch impedance along the path, a continuous symmetric representation embedded per head as a learnable bias. Since §3.2 guarantees strictly radial sub-zones, the path $\mathcal{P}(i,j)$ is unique and d_{elec} is well defined.

3.4.1. Electrical-distance matrix

The electrical distance between nodes i and j accumulates $|Z_{uv}|$ (§2.1) along the unique path of the radial tree:

$$d_{\mathrm{elec}}(i,j) = \sum_{(u,v)\in\mathscr{P}(i,j)} |Z_{uv}| \tag{6}$$

This symmetric, non-negative metric has zero diagonal and decreases with proximity; $\mathbf{D}_{\mathrm{elec}} \in \mathbb{R}^{N\times N}$ is precomputed offline.

3.4.2. Nodal impedance-environment embedding

For node i, the mean and maximum of the features $[r_{ij}, x_{ij}, |Z_{ij}|, r_{ij}/x_{ij}]$ of its incident branches form a static descriptor $\mathbf{z}_i \in \mathbb{R}^8$, whose projection $\mathrm{MLP}(\mathbf{z}_i)$ enters the attention projections of §3.4.3.

3.4.3. Multi-head spatial attention with impedance bias

Every node is mapped by three per-head projections, $(\mathbf{Q}_{h,i},\mathbf{K}_{h,i},\mathbf{V}_{h,i}) = \mathbf{W}_h^{\mathrm{Q,K,V}}(\mathbf{h}'_{t,i} + \mathrm{MLP}(\mathbf{z}_i))$, so the impedance environment enters query, key and value alike; d_{elec} enters the attention score of head h:

$$\mathrm{Attn}_h(i,j) = \mathrm{softmax}_j\left(\frac{\mathbf{Q}_{h,i}^{\top}\mathbf{K}_{h,j}}{\sqrt{d_k}} - \gamma_h \frac{d_{\mathrm{elec}}(i,j)}{\lambda_h} + \beta_h\right) \tag{7}$$

where d_k is the per-head key dimension. The three bias scalars are learned with the network: γ_h (initialized to 0.5) sets the distance penalty, suppressing electrically remote pairs; λ_h (initialized to 1.0, kept positive by $|\cdot|+10^{-6}$) is a learnable spatial scale, so the heads work at local and whole-zone scales; and β_h (initialized to 0) offsets the baseline. The heads are then aggregated and written back to the layer input by one learnable residual weight:

$$\mathbf{h}''_{t,i} = \mathrm{LayerNorm}\left(\mathbf{h}'_{t,i} + \alpha_a\, \mathbf{W}_O \left[\Big\|_{h=1}^{n_h} \sum_{j\in\mathscr{V}_a} \mathrm{Attn}_h(i,j)\, \mathbf{V}_{h,j}\right]\right) \tag{8}$$

where $\|$ concatenates the n_h heads, $\mathbf{W}_O$ is the output projection and $\alpha_a = \mathrm{Sigmoid}(a_a) \in (0,1)$ weighs that branch against the residual term $\mathbf{h}'_{t,i}$, so a node can fall back on its own representation; the time mark of Eq. (1) is added before temporal encoding.

3.5. Time-frequency enhancement

Frequency-domain enhancement. Following FEDformer [33], the hidden representation is transformed by a real-valued FFT, the lowest $K_f = 8$ components are modulated by learnable complex weights $\mathbf{W}_r + j\mathbf{W}_i$, and an inverse transform with a residual connection returns to the time domain (j is the imaginary unit). **Multi-scale patch attention.** Following PatchTST [34], the series is divided into non-overlapping patches of sizes $p_1, p_2 \in \{1,2\}$; each branch performs "fold-project-self-attend-unfold" independently, and the $n_p = 2$ branches are fused by learnable softmax weights:

$$\mathbf{h}^{\mathrm{patch}} = \sum_{s=1}^{n_p} w_s \cdot \mathrm{PatchBranch}_s(\mathbf{h}) \tag{9}$$

$$\mathbf{w} = (w_1, \ldots, w_{n_p}) = \mathrm{softmax}(\mathbf{a}_p) \tag{10}$$

where $\mathrm{PatchBranch}_s(\cdot)$ is the branch of patch size p_s, w_s the s-th element of the normalized weight vector $\mathbf{w}$, and $\mathbf{a}_p \in \mathbb{R}^{np}$ a learnable log-weight vector. Both act *per node* along time, so the hidden tensor is reshaped to $\mathbb{R}^{(B\cdot N)\times T\times H}$; the internal bottleneck equals H (=64), so both branches share one representation space.

3.6. BiGRU temporal encoding

The spatially enhanced sequence of each node passes an $n_l = 1$-layer bidirectional GRU of hidden dimension $H/2$ per direction, then a linear projection and a residual connection with layer normalization:

$$\mathbf{e}_{t,i} = \mathrm{LayerNorm}\left(\mathbf{h}_{t,i} + \mathrm{Dropout}\left(\mathbf{W}_{\mathrm{proj}}\left[\overrightarrow{\mathbf{h}}_{t,i} \,\|\, \overleftarrow{\mathbf{h}}_{t,i}\right]\right)\right) \tag{11}$$

where $\overrightarrow{\mathbf{h}}_{t,i}$ and $\overleftarrow{\mathbf{h}}_{t,i}$ are the forward and backward hidden states, $\mathbf{W}_{\mathrm{proj}}$ the projection weight, and Dropout(·) random dropout. Since the node dimension is folded into the batch dimension, the weights are shared across nodes and the parameter count $\mathcal{O}(n_l H^2)$ is independent of N, preserving dimension decoupling.

3.7. DHC hard-constraint output head

3.7.1. Physical roles of the output variables

DHC splits the zone outputs by whether a variable is predicted or inferred. The *free variables* $\mathscr{T}_{\mathrm{free}}$ are predicted directly: the root voltage V_{root}, the injections $\{P_i, Q_i\}_{i\in va}$, and the sending-end flows of the tie branches leaving the zone $\{P_{f,\mathrm{out},e}, Q_{f,\mathrm{out},e}\}_{e\in\mathrm{out}(va)}$, one pair per outgoing branch e, which characterize the exchange with neighboring zones. The *dependent variables* $\mathscr{T}_{\mathrm{dep}}$, namely the non-root voltages $\{V_k : k \neq \mathrm{root}\}$, root-node incident tie flows $(P_{f,\mathrm{entry}}, Q_{f,\mathrm{entry}})$, and squared branch currents $\{\ell_{ik}\}$, follow uniquely from the recursion once $\mathscr{T}_{\mathrm{free}}$ and the impedances are given. The tie-switch SCADA data serve as input features and soft-supervision targets, not as hard boundaries, and the current magnitude $|I_f|$ stays outside the recursion: an auxiliary head regresses it and it joins the analytical channels in $\hat{\mathbf{y}}$.

3.7.2. Free-variable prediction and temporal denoising

Three per-node shared heads map the encoder output $\mathbf{e} \in \mathbb{R}^{N\times H}$ to the free quantities, concatenated by channel into $\hat{\mathbf{y}}_t^{\mathrm{f}} \in \mathbb{R}^{|\mathscr{T}_{\mathrm{free}}|}$, superscripts f and d abbreviating $\mathscr{T}_{\mathrm{free}}$ and $\mathscr{T}_{\mathrm{dep}}$, not the subscript f of a tie branch, with $|\mathscr{T}_{\mathrm{free}}| = 1 + 2N + 2|\mathrm{out}(\mathscr{V}_a)|$:

$$\left(\widehat{P}_{t,i}, \widehat{Q}_{t,i}\right)^{\top} = \mathbf{W}_f \mathbf{e}_{t,i} + \mathbf{b}_f, i \in \mathscr{V}_a \tag{12}$$

$$\widehat{V}_{t,\mathrm{root}} = \mathbf{w}_V^{\top}\mathbf{e}_{t,\mathrm{root}} + b_V \tag{13}$$

$$\left(\widehat{P}_{f,\mathrm{out},e}, \widehat{Q}_{f,\mathrm{out},e}\right)^{\top} = \mathbf{W}_{\mathrm{out}}\mathbf{e}_{t,i(e)} + \mathbf{b}_{\mathrm{out}}, e \in \mathrm{out}(\mathscr{V}_a) \tag{14}$$

where $\mathbf{W}_f \in \mathbb{R}^{2\times H}$, $\mathbf{w}_V \in \mathbb{R}^H$ and $\mathbf{W}_{\mathrm{out}} \in \mathbb{R}^{2\times H}$ are the projection weights, $\mathbf{b}_f$, b_V and $\mathbf{b}_{\mathrm{out}}$ the corresponding biases, and $i(e)$ the in-zone starting node of the outgoing branch e. The three heads share weights across nodes or outgoing branches, so the parameter complexity is $\mathcal{O}(H)$, independent of N as in §3.2.2.

Free-variable noise is amplified twice: power accumulation builds variance with subtree scale, and voltage sweeping builds subtraction error with tree depth. A depthwise 1-D convolution [35], one kernel per channel and no mixing across channels, therefore smooths the window sequence $\hat{\mathbf{y}}_{1:T}^{\mathrm{f}}$ along time before the recursion, $\mathbf{s}_t = \mathrm{DWConv1D}\left(\hat{\mathbf{y}}_{1:T}^{\mathrm{f}}\right)_t$ being its output at step t. A learnable residual mix follows:

$$\tilde{\mathbf{y}}_t^{\mathrm{f}} = (1 - \boldsymbol{\alpha}_d)\, \hat{\mathbf{y}}_t^{\mathrm{f}} + \boldsymbol{\alpha}_d\, \mathbf{s}_t \tag{15}$$

where $\boldsymbol{\alpha}_d = \mathrm{Sigmoid}(\mathbf{a}_d) \in (0,1)^{|\mathscr{T}_{\mathrm{free}}|}$ is a channel-wise learnable vector, $\mathbf{a}_d$ its unconstrained parameter, so that each measurement type learns its own denoising strength. $\tilde{\mathbf{y}}_t^{\mathrm{f}}$ replaces $\hat{\mathbf{y}}_t^{\mathrm{f}}$ in the recursion.

3.7.3. DistFlow equations

In a radial zone every non-root node k has a unique parent $i = \pi(k)$, forming branch $(i \to k)$. Let $\mathscr{C}(k)$ be the downstream node set of k; P_{km}, Q_{km} the sending-end flows of $(k \to m)$; P_{ik}, Q_{ik} those of $(i \to k)$ at the i-end; ℓ_{ik} the squared branch current; and P_k, Q_k the injections (positive for consumption). The Baran-Wu equations [28,36] read

$$\begin{cases} P_{ik} = P_k + \sum_{m\in\mathscr{C}(k)} P_{km} + r_{ik}\ell_{ik} \\ Q_{ik} = Q_k + \sum_{m\in\mathscr{C}(k)} Q_{km} + x_{ik}\ell_{ik} \end{cases} \tag{16}$$

$$V_k^2 = V_i^2 - 2(r_{ik}P_{ik} + x_{ik}Q_{ik}) + (r_{ik}^2 + x_{ik}^2)\ell_{ik} \tag{17}$$

$$\ell_{ik} = \frac{P_{ik}^2 + Q_{ik}^2}{V_i^2} \tag{18}$$

where r_{ik}, x_{ik} are the branch parameters (§2.1) and V_i, V_k the terminal voltages. Eq. (16) is the power balance at node k (inflow = local consumption + downstream injections + loss), Eq. (17) the voltage-square drop along ($i \to k$), and Eq. (18) follows from $\ell_{ik} = |I_{ik}|^2$ with $S_{ik} = V_i I_{ik}^*$, where I_{ik} and S_{ik} are the sending-end current and complex power of the branch.

To treat the zone boundary uniformly, the *effective injection* of node k adds all tie flows leaving that node, which are equivalent to additional loads:

$$\begin{cases} \widetilde{P}_k \triangleq P_k + \sum_{e \in \mathrm{out}(k)} P_{f,\mathrm{out},e} \\ \widetilde{Q}_k \triangleq Q_k + \sum_{e \in \mathrm{out}(k)} Q_{f,\mathrm{out},e} \end{cases} \tag{19}$$

For a non-outgoing node, $\mathrm{out}(k) = \varnothing$ and the definition degenerates to $\widetilde{P}_k = P_k$, $\widetilde{Q}_k = Q_k$; the recursion therefore replaces P_k, Q_k in Eq. (16) by $\widetilde{P}_k$, $\widetilde{Q}_k$ without distinguishing boundary from interior nodes. Eq. (19) adds no constraint and merely extends DistFlow consistently at the boundary.

3.7.4. Manifold-constrained optimization and the differentiable DistFlow operator

DHC embeds Eqs. (16)-(19) in a differentiable analytical operator $f_{\mathrm{DF}}: \mathbf{y}^{\mathrm{f}} \to \mathbf{y}^{\mathrm{d}}$, turning training into manifold-constrained optimization:

$$\min_{\boldsymbol{\theta}} \mathscr{L}(\hat{\mathbf{y}}(\boldsymbol{\theta}); \mathbf{y}^*) \text{ s.t. } \hat{\mathbf{y}}^{\mathrm{d}} = f_{\mathrm{DF}}\left(\hat{\mathbf{y}}^{\mathrm{f}}(\boldsymbol{\theta}); \mathbf{r}, \mathbf{x}\right) \tag{20}$$

where $\boldsymbol{\theta}$ are the learnable parameters, $\mathscr{L}$ the loss, $\hat{\mathbf{y}}(\theta) = (\hat{\mathbf{y}}^{\mathrm{f}}, \hat{\mathbf{y}}^{\mathrm{d}})$ the complete output, $\mathbf{y}^*$ the ground truth, and $\mathbf{r}$, $\mathbf{x}$ the branch resistance and reactance vectors. The constraint means the dependent variables are never optimized independently but always follow from f_{DF}, so physical consistency is embedded in the feasible region.

The admissible outputs form a low-dimensional DistFlow feasible manifold $\mathscr{M}_{\mathrm{DF}}$, to which DHC structurally confines the output. f_{DF} consists of a leaf-to-root backward power accumulation and a root-to-leaf forward voltage sweep. The initial voltage $V^{(0)}$ comes from LinDistFlow ($\ell \equiv 0$) and the initial flows from the corresponding lossless accumulation. Round $n \geq 1$ first accumulates backward in order of decreasing depth, with $\widetilde{P}_k, \widetilde{Q}_k$ from Eq. (19):

$$\begin{cases} P_{ik}^{(n)} = \widetilde{P}_k + \sum_{m \in \mathscr{C}(k)} P_{km}^{(n)} + r_{ik} \widetilde{\ell}_{ik}^{(n)} \\ Q_{ik}^{(n)} = \widetilde{Q}_k + \sum_{m \in \mathscr{C}(k)} Q_{km}^{(n)} + x_{ik} \widetilde{\ell}_{ik}^{(n)} \end{cases} \tag{21}$$

The same expression gives the squared branch current at two rounds: the predictor $\widetilde{\ell}_{ik}^{(n)}$ uses the previous round, which avoids implicit coupling within a round, whereas the correction $\ell_{ik}^{(n)}$ uses the flows and parent voltage of the current round:

$$\begin{cases} \widetilde{\ell}_{ik}^{(n)} = \dfrac{\left(P_{ik}^{(n-1)}\right)^2 + \left(Q_{ik}^{(n-1)}\right)^2}{\left(V_i^{(n-1)}\right)^2} \\ \ell_{ik}^{(n)} = \dfrac{\left(P_{ik}^{(n)}\right)^2 + \left(Q_{ik}^{(n)}\right)^2}{\left(V_i^{(n)}\right)^2} \end{cases} \tag{22}$$

The voltage is then swept forward in order of increasing depth, with $V_{\mathrm{root}}^{(n)} = V_{\mathrm{root}}$:

$$\left(V_k^{(n)}\right)^2 = \left(V_i^{(n)}\right)^2 - 2\left(r_{ik} P_{ik}^{(n)} + x_{ik} Q_{ik}^{(n)}\right) + \left(r_{ik}^2 + x_{ik}^2\right) \ell_{ik}^{(n)} \tag{23}$$

After ν rounds we take $V_k = V_k^{(\nu)}$; both cases use $\nu = 2$, since the Baran-Wu sweep already reaches the 10^{-5} p.u. violation magnitude after two rounds. The incident tie flows $P_{f,\mathrm{entry}}, Q_{f,\mathrm{entry}}$ follow from the accumulation of Eq. (21) at the root node and need no separate prediction. Composed of elementary arithmetic and square roots, f_{DF} is smoothly differentiable everywhere.

3.7.5. Loss-gradient backpropagation

Because f_{DF} is smoothly differentiable, PIGAT trains end-to-end by gradients. The loss supervises $\hat{\mathbf{y}}^{\mathrm{f}}$ directly and also indirectly through $\hat{\mathbf{y}}^{\mathrm{d}} = f_{\mathrm{DF}}(\hat{\mathbf{y}}^{\mathrm{f}})$; by the chain rule the dependent-variable contribution is backpropagated by the Jacobian transpose:

$$\nabla_{\hat{\mathbf{y}}^{\mathrm{f}}} \mathscr{L} = \nabla^{\mathrm{dir}}_{\hat{\mathbf{y}}^{\mathrm{f}}} \mathscr{L} + \mathbf{J}_{f_{\mathrm{DF}}}^{\top} \nabla_{\hat{\mathbf{y}}^{\mathrm{d}}} \mathscr{L} \tag{24}$$

where $\mathbf{J}_{f\mathrm{DF}} \triangleq \partial \hat{\mathbf{y}}^{\mathrm{d}} / \partial \hat{\mathbf{y}}^{\mathrm{f}}$ is the Jacobian defined by Eqs. (21)-(23) and ∇^{dir} the direct gradient for fixed $\hat{\mathbf{y}}^{\mathrm{d}}$. Expanding Eq. (24) for a free component $\varphi \in \mathscr{F}_{\mathrm{free}}$ over the three types of dependent channels gives

$$\frac{\partial \mathscr{L}}{\partial \varphi} = \frac{\partial \mathscr{L}}{\partial \varphi}\Big|_{\mathrm{dir}} + \sum_{k \in \mathscr{V}_a, k \neq \mathrm{root}} \frac{\partial \mathscr{L}}{\partial \widehat{V}_k} \frac{\partial \widehat{V}_k}{\partial \varphi} + \sum_{\psi \in \{\hat{P}_{f,\mathrm{entry}}, \hat{Q}_{f,\mathrm{entry}}\}} \frac{\partial \mathscr{L}}{\partial \psi} \frac{\partial \psi}{\partial \varphi} + \sum_{(i,k) \in \mathscr{E}_a} \frac{\partial \mathscr{L}}{\partial \widehat{\ell}_{ik}} \frac{\partial \widehat{\ell}_{ik}}{\partial \varphi} \tag{25}$$

where ψ runs over the two incident tie flows at the root node. The first term is the own-measurement residual of φ; the others are the voltage, incident-flow and current-square residuals backpropagated through $\mathbf{J}_{f\mathrm{DF}}^{\mathrm{T}}$. The Jacobian is never built explicitly: reverse accumulation along the graph of Eqs. (21)-(23) is equivalent to Eq. (24).

3.8. Loss Function

Since DHC enforces physical consistency by construction, no physical penalty term is needed and the objective is a single Huber loss:

$$\mathscr{L} = \frac{1}{n_s} \sum_{t,d} \rho_\delta\left(\hat{y}_{t,d} - y_{t,d}\right), \rho_\delta(\xi) = \begin{cases} \frac{1}{2}\xi^2, & |\xi| \leq \delta \\ \delta|\xi| - \frac{1}{2}\delta^2, & |\xi| > \delta \end{cases} \tag{26}$$

where $\hat{y}_{t,d}$ and $y_{t,d}$ are the prediction and ground truth of feature d at step t, ξ the residual of the Huber kernel ρ_δ, δ its transition threshold (1.0), and n_s the number of supervised entries. The loss supervises the free and the dependent variables simultaneously, automatic differentiation returning the dependent-variable error to the free variables through Eqs. (24)-(25) and realizing the mechanism of §3.1.

4. Numerical experiments

4.1. Experimental setup

(1) **Datasets.** Two networks are used: Case 1 (IEEE 33-bus) and Case 2 (IEEE 123-bus). The weakly meshed IEEE 123-bus system is reduced to five independent radial sub-zones by cutting at tie switches, so that PIAB's "cumulative impedance along the unique path" metric holds strictly inside each sub-zone; Case 1 is divided into five zones likewise. Ordinary load profiles come from the public UCI ElectricityLoadDiagrams20112014 dataset [37], scaled to a 20% load share. Each window sample contains 16 consecutive 15-min intervals (4 h): the first 8 are backfilled complete historical AMI data, the last 8 (the most recent 2 h) the target interval with missing values; samples are split chronologically into 70% training and 30% test.

(2) **Scenario simulation.** Three scenario families are designed. *Missing rate:* random missingness of 20%, 40%, and 60% is injected into the AMI measurements of the last 8 intervals of each window. *Noise:* measurement noise is modeled as additive zero-mean Gaussian error, i.e. an observation y is perturbed to $y + \varepsilon$ with $\varepsilon \sim \mathscr{N}(0, \sigma^2)$, the standard deviation σ of a channel being

Table II
Zonal pseudo-measurement imputation results for Case 1 and Case 2 (60% missing).

Data	Model	Per-zone RMSE / s per epoch / ms per sample					Five-zone average RMSE / s/ep / ms
		Area 1	Area 2	Area 3	Area 4	Area 5	
Case 1	MF	0.0072/0.034/0.064	0.0209/0.118/0.194	0.0123/0.226/0.183	0.0249/0.172/0.132	0.0230/0.099/0.179	0.0177/0.130/0.150
	GAIN	0.0083/0.468/0.005	0.0198/0.483/0.005	0.0112/0.493/0.005	0.0214/0.482/0.005	0.0197/0.482/0.005	0.0161/0.482/0.005
	GRIN	0.0094/3.728/0.013	0.0184/3.760/0.019	0.0101/3.769/0.018	0.0233/3.790/0.016	0.0240/3.784/0.018	0.0171/3.766/0.017
	SAITS	0.0040/0.303/0.002	0.0092/0.322/0.002	0.0061/0.314/0.002	0.0095/0.309/0.002	0.0085/0.304/0.002	0.0075/0.310/0.002
	PIGAT	**0.0033**/1.185/0.022	**0.0078**/2.474/0.045	**0.0050**/2.063/0.036	**0.0083**/2.002/0.037	**0.0082**/2.226/0.040	**0.0065**/1.990/0.036
Case 2	MF	0.0234/1.285/0.873	0.0190/0.550/0.461	0.0175/0.555/0.604	0.0224/1.718/1.270	0.0265/0.684/0.636	0.0218/0.958/0.769
	GAIN	0.0232/0.508/0.001	0.0244/0.486/0.001	0.0188/0.520/0.001	0.0264/0.557/0.005	0.0244/0.522/0.001	0.0234/0.519/0.002
	GRIN	0.0291/3.862/0.423	0.0389/3.579/0.048	0.0164/3.650/0.060	0.0321/3.809/0.792	0.0356/3.863/0.058	0.0304/3.753/0.276
	SAITS	0.0099/0.360/0.002	0.0084/0.339/0.002	0.0080/0.328/0.001	0.0109/0.365/0.002	0.0094/0.334/0.002	0.0093/0.345/0.002
	PIGAT	**0.0075**/6.051/0.121	**0.0075**/3.414/0.059	**0.0062**/3.601/0.064	**0.0094**/6.770/0.131	**0.0078**/4.163/0.083	**0.0077**/4.800/0.092

fixed at the stated percentage of its nominal full-scale value. Two accuracy classes are distinguished because voltage channels are metered more accurately than power channels: σ_V applies to voltage magnitudes and σ_S to branch flows, branch currents, and nodal injections. Scenario 1 (σ_V=0.5%, σ_S=1%) represents gateway metering with class-0.5 instrument transformers; Scenario 2 (1%, 2%) represents field-deployed AMI and feeder terminal units and is the default of all other experiments; Scenario 3 (2%, 3%) represents aged or lower-class metering together with communication-induced quantization error. Noise is drawn independently for every sample, channel, and time step, and is applied to the model input only, the noise-free values serving as evaluation ground truth. All perturbed datasets are generated once under a fixed seed, so the five models see exactly the same corrupted data. *Penetration:* the total DER generation is set to 30%, 50%, and 70% of the positive load; photovoltaic output follows a daytime sinusoidal pattern modulated by random cloud shading, wind power superimposes random disturbances on a diurnal baseline, and EV charging follows an intraday time-period profile.

(3) **Evaluation metrics.** Four metrics, MAE, RMSE, MAPE, and the coefficient of determination R^2, evaluate imputation accuracy:

$$\mathrm{MAE} = \frac{1}{n_s}\sum_{i=1}^{n_s}|y_i - \hat{y}_i| \tag{27}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n_s}\sum_{i=1}^{n_s}(y_i - \hat{y}_i)^2} \tag{28}$$

$$\mathrm{MAPE} = \frac{100\%}{n_s}\sum_{i=1}^{n_s}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \tag{29}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n_s}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n_s}(y_i - \bar{y})^2} \tag{30}$$

where y_i and $\hat{y}_i$ are the ground truth and prediction of sample i, $\bar{y}$ the mean of the ground truth, and n_s the sample count. All four adopt the *macro per-feature average*: computed per feature column, averaged over the columns of a zone and over the five zones, so dimensional differences are removed and every feature contributes equally.

The PVD quantifies the DistFlow residual by a power and a voltage term:

$$\mathrm{PVD}_P = \frac{1}{n_s}\sum_t\Big|\hat{P}_{f,t} - \sum_{i\in\mathscr{V}_a}\hat{P}_{i,t} - \sum_{\text{out}}\hat{P}_{f,t} - \sum_{(i,j)\in\mathscr{E}_a} r_{ij}\hat{\ell}_{ij,t}\Big| \tag{31}$$

$$\mathrm{PVD}_V = \frac{1}{|\mathscr{E}_a|\,n_s}\sum_{(i,j)}\sum_t\Big|\hat{V}_{i,t} - \sqrt{\hat{V}_{j,t}^2 - \left(r_{ij}^2 + x_{ij}^2\right)\hat{\ell}_{ij,t} + 2\left(r_{ij}\hat{P}_{ij,t} + x_{ij}\hat{Q}_{ij,t}\right)}\Big| \tag{32}$$

where $P_{f,t}$ and $P_{i,t}$ are the predicted entry tie-branch flow and nodal injection, $\Sigma_{\text{out}}P_{f,t}$ the sum of outgoing tie flows, $\hat{\ell}_{ij,t}$ the predicted squared branch current, $P_{ij,t}$ and $Q_{ij,t}$ the predicted sending-end powers, $V_{i,t}$ and $V_{j,t}$ the predicted terminal voltages, $\mathscr{V}_a$ and $\mathscr{E}_a$ the zone node and branch sets (§3.2), and r_{ij}, x_{ij} the branch parameters. The overall value is $\mathrm{PVD}_{\text{total}} = (\mathrm{PVD}_P + \mathrm{PVD}_V)/2$.

(4) **Model parameters and implementation platform.** The baselines GRIN [17], MissForest (MF) [13], GAIN [15], and SAITS [16] represent graph-neural-network, random-forest, adversarial, and self-attention imputation. All deep models share one configuration: Adam, learning rate 0.001, batch size 512, at most

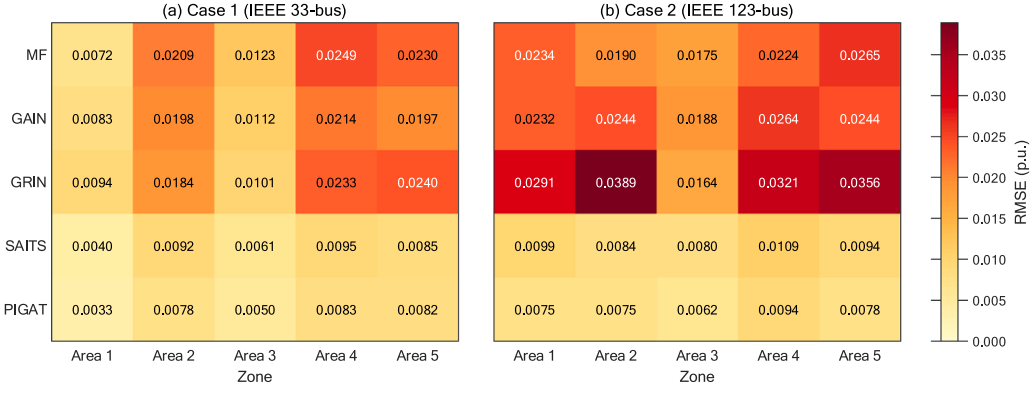


Fig. 4. Heat map of the zonal RMSE of Table II (60% missing).

Table III
Imputation performance under different missing rates (five-zone average).

Data	Model	20% missing				40% missing				60% missing			
		R^2	MAE	RMSE	MAPE(%)	R^2	MAE	RMSE	MAPE(%)	R^2	MAE	RMSE	MAPE(%)
Case 1	MF	0.970	0.0049	0.0117	5.93	0.951	0.0074	0.0146	8.70	0.924	0.0101	0.0177	11.60
	GAIN	0.947	0.0094	0.0134	10.54	0.938	0.0098	0.0143	10.47	0.917	0.0110	0.0161	11.67
	GRIN	0.969	0.0047	0.0104	4.83	0.940	0.0075	0.0140	8.63	0.906	0.0104	0.0171	11.45
	SAITS	0.994	0.0029	0.0045	3.21	0.990	0.0036	0.0057	4.36	0.983	0.0049	0.0075	5.81
	PIGAT	**0.995**	**0.0023**	**0.0040**	**2.42**	**0.992**	**0.0030**	**0.0050**	**3.45**	**0.987**	**0.0040**	**0.0065**	**4.87**
Case 2	MF	0.948	0.0053	0.0120	3.27	0.897	0.0086	0.0171	5.50	0.841	0.0123	0.0218	8.33
	GAIN	0.892	0.0130	0.0182	8.71	0.857	0.0144	0.0204	9.02	0.803	0.0165	0.0234	9.49
	GRIN	0.946	0.0125	0.0196	4.00	0.901	0.0166	0.0249	6.48	0.838	0.0213	0.0304	8.91
	SAITS	0.977	0.0045	0.0066	3.06	0.966	0.0052	0.0078	3.76	0.952	0.0063	0.0093	4.72
	PIGAT	**0.990**	**0.0031**	**0.0053**	**1.98**	**0.983**	**0.0041**	**0.0064**	**2.86**	**0.973**	**0.0051**	**0.0077**	**3.80**

200 epochs, early stopping with patience 30 and a minimum improvement of 10^{-7}, dropout 0.1, and the Huber loss (δ=1.0) except GAIN, which is trained adversarially. Method-specific settings are 5 imputation rounds with 100 trees of depth 6 for MF, a hint rate of 0.9 and a reconstruction weight of 100 for GAIN, and two encoder blocks with two heads for SAITS; PIGAT uses H= 64, n_h= 2, n_l= 1, K_f= 8, patch sizes {1, 2} and ν= 2 sweep rounds in DHC. All results are means of five independent runs, obtained on an 11th-generation Intel Core i7–11700 CPU at 2.50 GHz with one NVIDIA GeForce RTX 3080 GPU (10 GB), under Python 3.10.9, PyTorch 2.5.1 and scikit-learn 1.2.1 with CUDA 12.4, MF running on the CPU, so that all reported times are measured in a single environment.

4.2. Zonal pseudo-measurement imputation comparison

With 60% missing, noise Scenario 2, and 30% DER, Table II reports the RMSE, training time (s/epoch), and test time (ms/sample) of Areas 1–5, the rightmost column being the five-zone average.

Table II and Fig. 4 compare the five methods per zone. PIGAT and SAITS take the top two RMSE positions in both cases, PIGAT always ahead; their five-zone averages, 0.0065 and 0.0077, are 2–4 × lower than MF, GAIN, and GRIN, whose own ranking varies with scale (Case 1: GAIN < GRIN < MF, 0.0161 < 0.0171 < 0.0177; Case 2: MF < GAIN < GRIN, 0.0218 < 0.0234 < 0.0304). GRIN, the only graph-convolutional baseline, falls last in Case 2 and degrades further with scale because it aggregates over the graph *before* correcting the missing entries, so placeholder zeros are carried to the neighbors of every gap and this *missing-value contamination* grows with scale and coupling. SAITS, which does not propagate spatially, still ranks second steadily: graph convolution pays off only when the order is reversed, as in PIGAT, which corrects first and propagates afterwards inside a small sub-zone. Partitioned parallel training gives about 4.0 × speedup in Case 1 (slowest zone 2.47 vs 9.95 s/epoch sequentially) and 3.5 × in Case 2 (6.77 vs 24.00 s/epoch). Fig. 4 adds what the table cannot show at a glance: pale and dark columns repeat down the rows, so zone difficulty is mainly a property of the sub-network rather than of the model. Area 1 in Case 1 and Area 3 in Case 2, with the fewest nodes and the shortest feeders, are the lightest cells of every row, while Areas 4 and 5 stay dark and the two bottom rows are uniformly pale. The single exception, GRIN in Area 2 of Case 2, is the darkest cell of a zone that is easy for the other four models, so that error follows the model rather than the network.

4.3. Robustness under different missing rates

With noise Scenario 2 and 30% DER fixed, the missing rates 20%/40%/60% are evaluated; under each rate the five sub-zones are trained independently and the metrics averaged. Table III reports the five-zone averages.

All models degrade as the missing rate rises from 20% to 60%, but at markedly different rates. PIGAT degrades least (R^2 falls by only 0.008 and 0.017), the data-driven baselines most (GRIN by 0.063 and 0.108, MF by 0.046 and 0.107); at 60% missing its RMSE is 13%-17% better than the runner-up SAITS and up to 2.78 × better than MF. Two pathways explain this: PIAB borrows valid observations from electrically nearby nodes along the impedance-weighted path, and the BiGRU with mask gating trusts the observed entries. GRIN collapses because more placeholder zeros mean more contaminated neighbors once it aggregates before correcting (§4.2), and MF degrades because it treats each variable independently. Fig. 5 makes the divergence visible. Read left to right, the two attention rows darken slowly while the other three darken sharply, so the gap between PIGAT and the three non-attention baselines widens from 0.0064 to 0.0143 at 20% missing to 0.0096–0.0227 at 60%, precisely in the regime AMI latency creates. Read diagonally, the PIGAT cell at 60% missing is lighter than every MF, GAIN, and GRIN cell at 20%: PIGAT tolerates a missing rate forty points higher and still returns

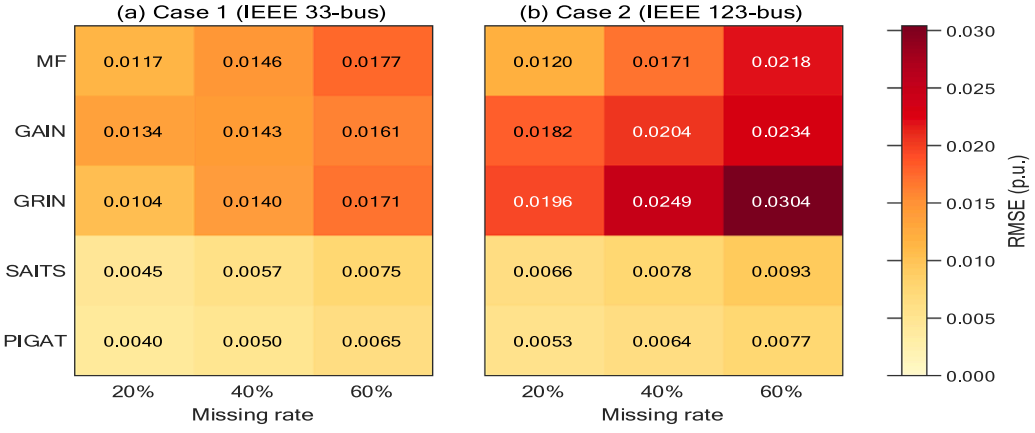


Fig. 5. Heat map of the five-zone average RMSE of Table III against the missing rate.

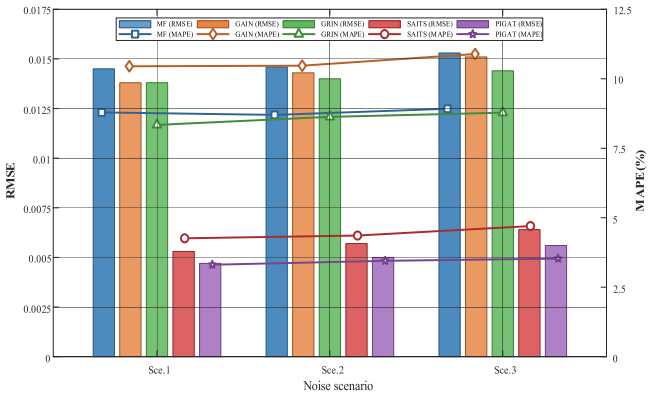


Fig. 6. Imputation performance under different noise scenarios (Case 1).

the lower RMSE; only SAITS stays lighter at 20%.

4.4. *Adaptability to noise and DER penetration*

(1) **Noise scenarios.** With 40% missing and 30% DER, Figs. 6 and 7 give the five-zone average RMSE (bars, left axis) and MAPE (lines, right axis) under the three noise scenarios.

PIGAT gives the lowest RMSE and MAPE under all three scenarios. Under severe noise (Scenario 3) its RMSE is 0.0056 and 0.0074, about 13% below the runner-up SAITS with a MAPE 0.67 points lower, and up to 67% below GAIN. From Scenario 1 to Scenario 3 its relative RMSE increase (19.1%-25.4%) slightly exceeds that of SAITS (16.4%), but its much lower starting point keeps the absolute deterioration ($\leq$0.0015) within the best region. The robustness comes from the low-pass effect of impedance-weighted attention on spatially uncorrelated noise and from DHC, which confines the solution to the DistFlow manifold before power accumulation.

(2) **DER penetration.** With 40% missing and noise Scenario 2, DER penetration takes 30%/50%/70%; results are shown in Figs. 8 and 9.

The RMSE of all models grows monotonically from 30% to 70% penetration, reflecting the difficulty added by high-frequency renewable fluctuation. Most models show a local MAPE spike at 50% (e.g. MF in Case 2: 5.50→14.63→7.40), because this level coincides with the "load ≈ renewable output" zero-crossing that amplifies the denominator sensitivity of MAPE, so MAPE alone is not a degradation criterion here. PIGAT compresses that peak: its relative peak amplitude in Case 2 is + 30.1%, about 35%, 64%, and 82% lower than SAITS (+46.0%), GAIN (+84.3%), and MF (+166.0%). Its three-level RMSE increase is only + 8.0% and + 9.4%, well below GRIN (+45.0%, +32.9%) and MF (+15.8%, +15.2%); GAIN increases as little, but its absolute RMSE stays

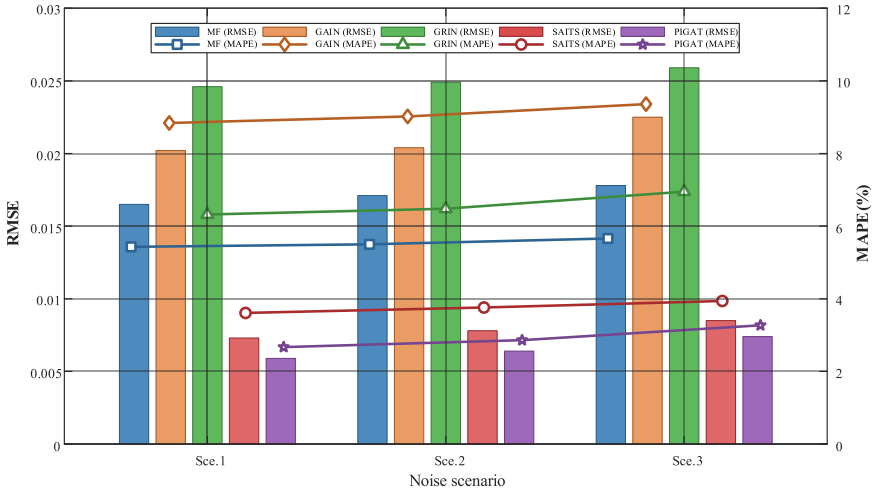


Fig. 7. Imputation performance under different noise scenarios (Case 2).

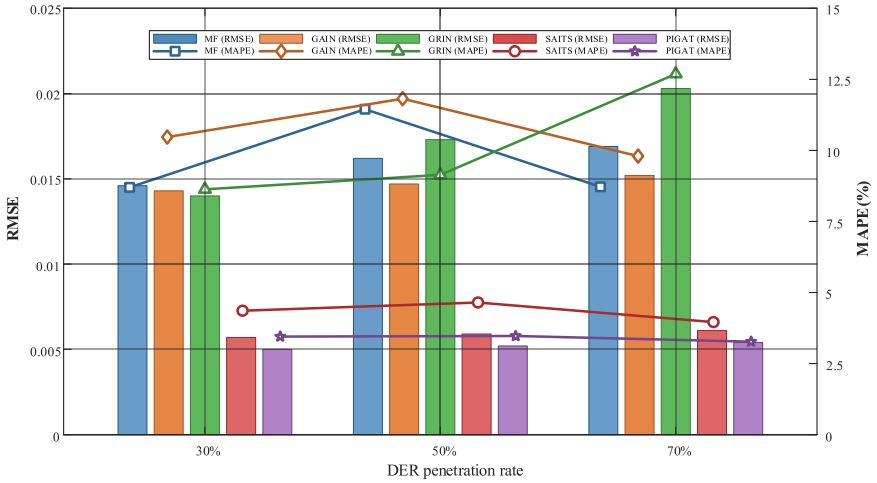


Fig. 8. Imputation performance under different DER penetrations (Case 1).

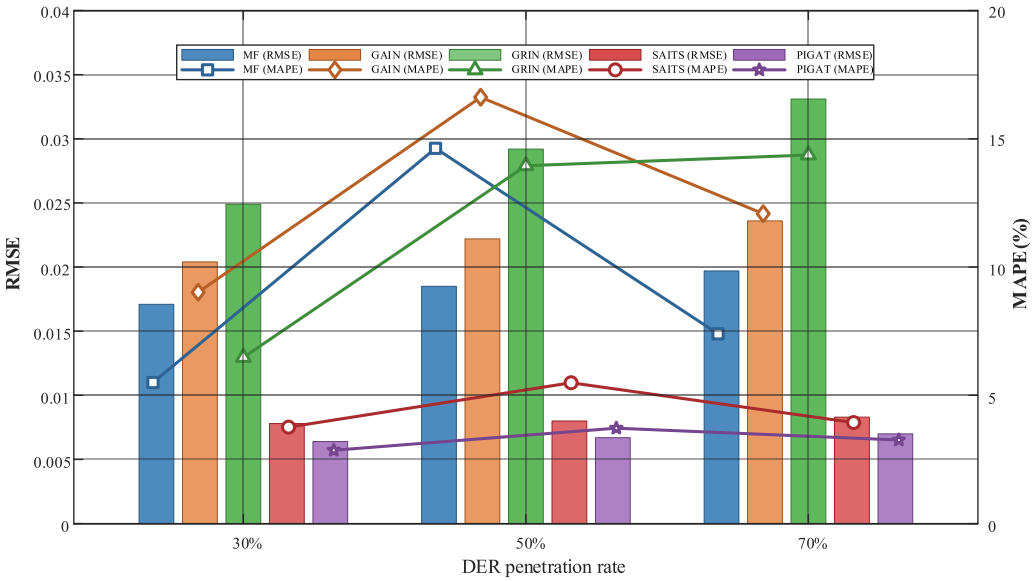


Fig. 9. Imputation performance under different DER penetrations (Case 2).

Table IV
Ablation study results.

Data	Model	MAE	RMSE	MAPE(%)	R^2	PVD_{total}
Case 1	**PIGAT**	**0.0023**	**0.0040**	**2.15**	**0.9937**	**3.1×10^{-5}**
	w/o Gate	0.0030	0.0049	3.10	0.9909	3.1×10^{-5}
	w/o PIAB	0.0024	0.0044	2.25	0.9924	3.1×10^{-5}
	w/o DHC	0.0025	0.0042	2.36	0.9932	1.2×10^{-3}
Case 2	**PIGAT**	**0.0034**	**0.0053**	**2.10**	**0.9864**	**4.6×10^{-5}**
	w/o Gate	0.0038	0.0060	2.58	0.9811	4.6×10^{-5}
	w/o PIAB	0.0036	0.0056	2.41	0.9849	4.6×10^{-5}
	w/o DHC	0.0036	0.0055	2.24	0.9853	1.6×10^{-3}

about 3 × that of PIGAT, so the small increase merely reflects a poor baseline. The robustness comes from PIAB raising the spatial weight of conventional load nodes when renewable-side variance grows, and from DHC excluding physically unreasonable estimates.

4.5. Ablation study

Table IV presents the ablation study results. Three variants isolate the components: (1) *w/o* Gate replaces the mask-aware gate by feature concatenation; (2) *w/o* PIAB falls back to a ChebNet-style graph convolution without impedance weighting; (3) *w/o* DHC trains with a purely data-driven Huber loss. The BiGRU backbone is retained throughout, and all ablations use Area 3 of each case, 40% missing, noise Scenario 2, and 30% DER.

Removing any core component degrades every accuracy metric in both cases, validating the three designs. The gate contributes most: its removal gives the largest single-component RMSE increase (+22.5% and +13.2%), corroborating the role of feature-level correction before spatial aggregation. Removing PIAB forfeits impedance heterogeneity and raises the RMSE by 10.0% and 5.7%, with a synchronous drop in R^2. DHC, the sole source of physical consistency, makes PVD_{total} jump from 10^{-5} to 10^{-3} (about 35–39 ×) when removed, while its manifold projection also brings RMSE gains of 5.0% and 3.8%. Every variant retaining DHC keeps its PVD at the 10^{-5} p.u. magnitude, so the physical guarantee is structurally decoupled from the gate and PIAB, a trade-off weighed in Section 5.

4.6. Physical consistency analysis

Table V compares the PVD ranges of the models.

Converted to actual values on the IEEE 33-bus base (S_{base}=10 MVA, V_{base}=12.66 kV), the worst-case power deviation of PIGAT is only 0.63 kW and its voltage deviation 0.63 V, about 0.005% of V_{base} and within 0.5% of the conservative 1%-of-V_{base} monitoring criterion adopted here, itself far stricter than the ±7% limit for medium-voltage three-phase supply specified in GB/T 12325–2008 [38]. Those of GAIN, GRIN, and SAITS reach 103.4 V, 92.8 V, and 63.0 V (0.5%-0.8% of V_{base}), an order of magnitude larger and already approaching that criterion. This exposes the difference between a *structural* guarantee and *incidental* plausibility: the dependent variables are derived analytically rather than learned, so the PVD of PIGAT stays at 10^{-5} p.u. under all data conditions, whereas the worst case of every baseline, the most accurate SAITS included, is one to two orders higher. High accuracy therefore does not imply physical consistency, and this residual lets PIGAT serve as a physically consistent front end for state estimation and Volt/VAR optimization.

5. Discussion

Three strengths stand out. PIGAT is the most accurate model in every scenario tested, and its margin is largest where AMI latency bites hardest: at 60% missingness its RMSE is 13%-17% below the runner-up SAITS and up to 2.78 × below MF (Table III). Its physical consistency

Table V
Overview of PVD comparison.

Model	PVD_{total} range	PVD_P range	PVD_V range	vs PIGAT (ratio of PVD_{total} upper bound)
PIGAT	1.2×10^{-6}~6.3×10^{-5}	1.5×10^{-6}~6.3×10^{-5}	8.4×10^{-7}~5.0×10^{-5}	Baseline
SAITS	2.1×10^{-4}~3.1×10^{-3}	6.6×10^{-5}~1.4×10^{-3}	1.9×10^{-4}~5.0×10^{-3}	~49 ×
MF	8.5×10^{-4}~5.2×10^{-3}	3.1×10^{-4}~7.8×10^{-3}	1.3×10^{-3}~5.3×10^{-3}	~83 ×
GAIN	1.4×10^{-3}~8.0×10^{-3}	3.8×10^{-4}~1.2×10^{-2}	2.2×10^{-3}~8.2×10^{-3}	~127 ×
GRIN	2.1×10^{-3}~1.0×10^{-2}	2.9×10^{-4}~1.3×10^{-2}	3.6×10^{-3}~7.3×10^{-3}	~159 ×

differs in kind, not degree: the dependent variables follow from an analytical recursion rather than from fitting, so the PVD stays at 10^{-5} p. u. whatever the data condition, 49–159 × below the baselines (Table V). Its scalability follows from the same design: the parameter count depends only on H, so one set of weights transfers across sub-zones of different sizes, the DHC recursion is linear in the zone size, and zone-parallel training is 3.5–4.0 × faster than sequential execution.

Three limitations qualify these results. The first is computational: PIGAT has the highest per-epoch training time of the five methods, 4.800 s against 0.345 s for SAITS in Case 2 (Table II), because the DistFlow sweep is inherently sequential in tree depth, and zone-parallel training relieves this by 3.5 × but cannot remove it. The burden is confined to offline training, which also fixes the intended deployment: a GPU-equipped distribution master-station server rather than computationally constrained embedded devices; online, one forward pass per sample costs 0.09–0.13 ms (Table II), far below the 15 min reporting period. The second is the reliance on boundary metering: accuracy and feasibility are obtained precisely by anchoring the recursion to the high-accuracy switch measurements that bound each zone, so a zone whose boundary switch is unmetered leaves the physical guarantee unsupported, and the partition must be drawn so that every zone root carries SCADA. The third is the dependence on model information: PIGAT needs a radial partition, a parent-child ordering and branch impedances where the baselines need only the measurement matrix, so erroneous impedances or undetected bad data bias the recursion, and the method is best applied downstream of parameter identification and bad-data detection; failing that, relaxing DHC into a soft penalty recovers part of the model independence at the price of strict feasibility.

These limitations set the future work. Dynamic topology reconfiguration will be addressed by updating the electrical-distance matrix and the recursion order incrementally after switching, so that the partition follows the operating configuration rather than a fixed one. Multi-resolution μPMU/AMI/SCADA fusion will relax the reliance on a single boundary source by combining streams of different accuracy and sampling rate under asynchronous time alignment. In parallel, the sensitivity of PIAB and DHC to impedance and topology errors will be quantified and the method combined with parameter identification and bad-data detection, together with the soft-constraint relaxation above; the DistFlow sweep will be parallelised or truncated with an adaptive number of rounds; and the imputation will be evaluated end-to-end through the state-estimation, Volt/VAR and demand-response decisions it supports [3].

6. Conclusion

For distribution-transformer-level pseudo-measurement imputation under AMI-latency-induced random missingness, this paper proposes PIGAT and validates it on the modified IEEE 33-/123-bus systems. Its three coupled mechanisms close the three research gaps along a "correct-then-propagate" pipeline. The mask-aware confidence gate contributes most to accuracy, its removal raising RMSE by 13.2%-22.5%, and pre-corrects missing entries before the PIAB impedance bias, which lowers RMSE by a further 5.7%-10.0% over a fixed-weight adjacency and confirms that electrical distance outperforms topological hops. The DHC head derives the dependent variables from an analytical DistFlow recursion instead of learning them, holding the PVD at 10^{-5} p.u. one to two orders below the baselines whatever the missing rate and noise. Partitioning with dimension decoupling ties the parameter count to H rather than N, so one set of weights transfers across radial networks of different scales and parallel training is 4.0 × and 3.5 × faster. Degradation is graceful: R^2 falls by only 0.008 and 0.017 across 20%-60% missing rates and RMSE rises by only 8.0% and 9.4% as DER penetration grows from 30% to 70%. The method therefore suits medium- and low-voltage radial active distribution networks with AMI and sparse boundary SCADA, as a physically consistent front end for state estimation, Volt/VAR optimization, and dispatch.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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